

Priyankkumar Chandrakant Patel

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TECHNICAL SKILLS

Fundamental Concepts	:Object Oriented Designs, Algorithm, Data Structures, Data Analysis, LLM.
Programming Languages	:Java, Python, C, JavaScript, R, SQL, Node.js, React.js.
Databases and Cloud	:MySQL, MongoDB, SQLite, GCP, AWS, Spark, PostgreSQL.
Libraries	:Pandas, Matplotlib, TensorFlow, Scikit Learn, NumPy, PyTorch.
Research Interests	:Language Modeling, RAG, Transformer Architecture, NLP, Reinforcement Learning.
Software Engineering	:System Design, Debugging, Testing, Code Reviews, Agile/Scrum.
Tools and Frameworks	:RESTful API, Tableau, Git, GitHub, CI/CD, FastAPI, Docker, LangChain.

WORK EXPERIENCE

AI Engineer Intern | Deepthink Healthcare | San Ramon, USA Sept 2025 - Dec 2025

- Built OCR preprocessing pipeline using Tesseract cues and Vision Transformers (ViT) to improve medical-document extraction quality and reduce manual review effort.
- Evaluated MedTech Q&A outputs through structured QA experiments, identifying failure patterns and improving reviewed accuracy to 90%.
- Developed transformer-embedding classification models for medical-device routing, achieving 95% F1 score in internal evaluations.
- Integrated AI model outputs into reproducible Python evaluation pipelines supporting automated routing and model benchmarking.
- Curated and annotated structured medical datasets in Label Studio to improve AI training quality and downstream model reliability.

ML Data Scientist | Enlighten Infosystems | Vadodara, India Jan 2023 - Aug 2023

- Performed EDA on yearly software sales to extract insights into the **company's past 5 years** of business strategies.
- Trained SVM models in Python on 3,000+ sales records, achieving **95% accuracy** for performance classification.
- Developed regression models on structured sales data, achieving **92% accuracy** for revenue prediction workflows.
- Applied clustering algorithms for customer segmentation, improving targeted marketing strategy and acquisition by **15%**.
- Preprocessed structured datasets, engineered features, and benchmarked ML models using Python, Scikit-learn, and evaluation metrics.

Software Development Intern | Elsner Technologies | Ahmedabad, India Apr 2022 - Aug 2022

- Developed iOS task-management features with calendar integration, improving **user engagement by 30%**.
- Optimized mobile app performance by **reducing load times 40%**, improving usability for large task lists.

PROJECTS

Pharmaceutical Sales Data Warehouse and Time Series Forecasting

- Built ETL pipeline for **200+ pharmaceutical sales datasets** using Snowflake schema and normalized relational design.
- Developed R dashboard with kableExtra to track sales performance across **20+ regions**.
- Constructed a data warehouse using OLAP principles to extract critical insights from over **200 datasets** on sales and performance.

Fake News Detection Using Machine Learning

- Experimented with RNN/LSTM sequence-model architectures on 10,000+ articles, improving classification performance by 25%.
- Applied NLP preprocessing, tokenization, and sequence modeling techniques to improve training efficiency and model generalization.
- Benchmarked model outputs using classification metrics and visualization-based error analysis.

Self Improving Conversational RAG System

- Built RAG system enabling conversational querying across multiple PDF documents using transformer embeddings.
- Evaluated chunking, retrieval, and prompt-engineering strategies to improve contextual grounding and reduce hallucinations.
- Developed FastAPI backend integrating semantic search, vector retrieval, and conversational AI workflows.
- Benchmarked retrieval quality and response consistency using embedding similarity and transformer-based evaluation metrics.

EDUCATION

Master of Science in Computer Science Jan 2024 - Present
Northeastern University, Boston, MA GPA: 3.8 / 4.0

Bachelor of Technology in Computer Science and Engineering Jun 2019 - May 2023
Gujarat Technological University, India GPA: 4.0 / 4.0

EXTRACURRICULAR ACTIVITY

- Masai Coding Teaching Assistant** : Guided **15 students** through Coding community for junior algorithm.
- Data Analytics Engineering** : Student Organization at Northeastern university, curation team member.
- Certifications** : Amazon AWS Cloud Architecting, Amazon AWS AML, Amazon Data Engineer.